

Assignment 3

1 Problem 1

With the given loss function, we can draw the following computational graph:

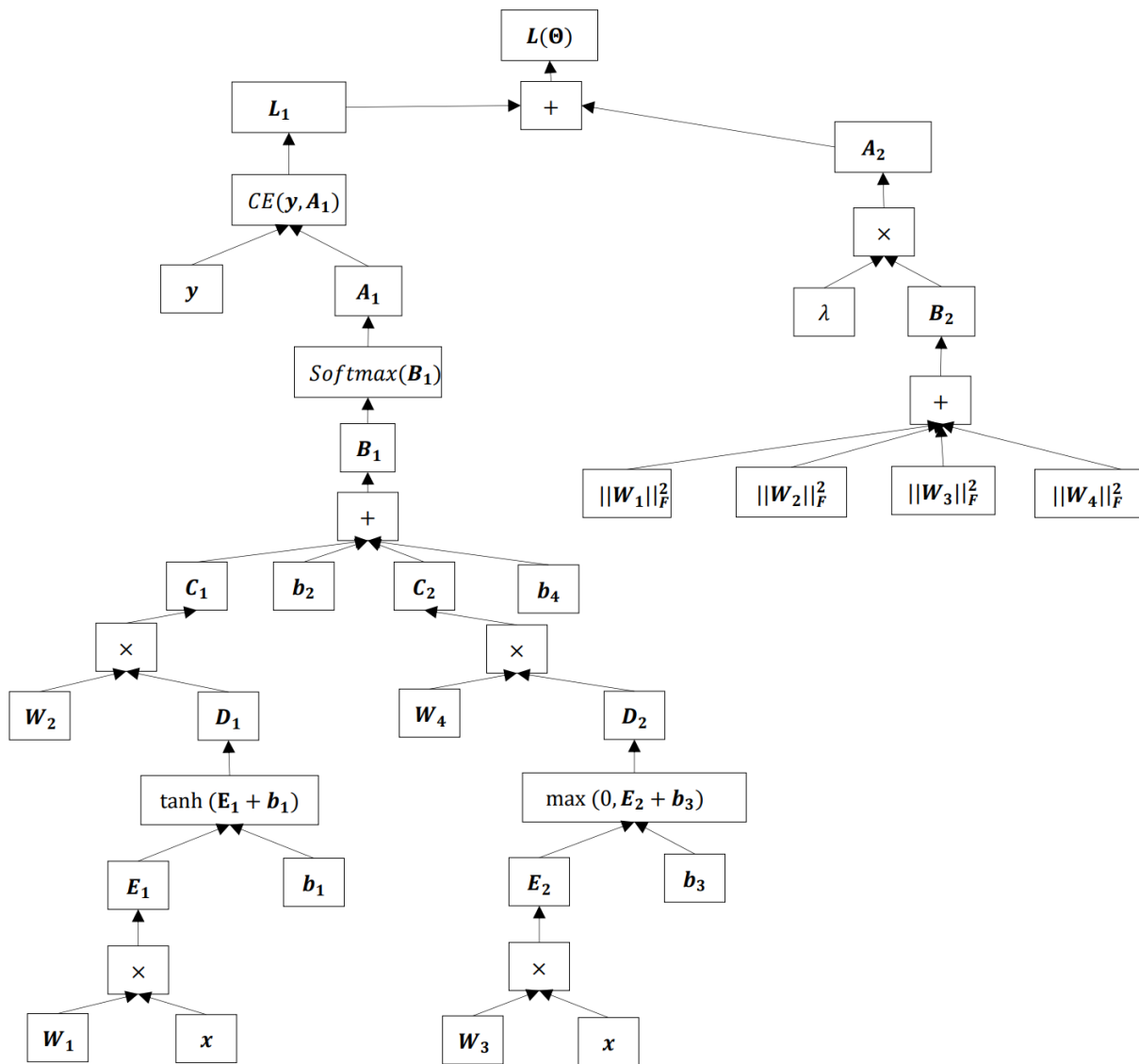


Figure 1: Computational Graph

The back-propagation includes two steps: forward pass and backward pass;
Forward pass:

$$\begin{aligned}
\mathbf{L}(\boldsymbol{\Theta}) &= \mathbf{L}_1 + \mathbf{A}_2 \\
\mathbf{L}_1 &= CE(\mathbf{y}, \mathbf{A}_1) \\
\mathbf{A}_1 &= \text{softmax}(\mathbf{B}_1) \\
\mathbf{B}_1 &= \mathbf{C}_1 + \mathbf{b}_2 + \mathbf{C}_2 + \mathbf{b}_4 \\
\mathbf{C}_1 &= \mathbf{W}_2 \mathbf{D}_1 \\
\mathbf{D}_1 &= \tanh(\mathbf{E}_1 + \mathbf{b}_1) \\
\mathbf{E}_1 &= \mathbf{W}_1 \mathbf{x} \\
\mathbf{C}_2 &= \mathbf{W}_4 \mathbf{D}_2 \\
\mathbf{D}_2 &= \max(0, \mathbf{E}_2 + \mathbf{b}_3) \\
\mathbf{E}_2 &= \mathbf{W}_3 \mathbf{x} \\
\mathbf{A}_2 &= \lambda \mathbf{B}_2 \\
\mathbf{B}_2 &= \|\mathbf{W}_1\|_{\text{F}}^2 + \|\mathbf{W}_2\|_{\text{F}}^2 + \|\mathbf{W}_3\|_{\text{F}}^2 + \|\mathbf{W}_4\|_{\text{F}}^2
\end{aligned}$$

Backward pass:

$$\begin{aligned}
\frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{W}_1} &= \frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{L}_1} \frac{\partial \mathbf{L}_1}{\partial \mathbf{A}_1} \frac{\partial \mathbf{A}_1}{\partial \mathbf{B}_1} \frac{\partial \mathbf{B}_1}{\partial \mathbf{C}_1} \frac{\partial \mathbf{C}_1}{\partial \mathbf{D}_1} \frac{\partial \mathbf{D}_1}{\partial \mathbf{E}_1} \frac{\partial \mathbf{E}_1}{\partial \mathbf{W}_1} + \frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{A}_2} \frac{\partial \mathbf{A}_2}{\partial \mathbf{B}_2} \frac{\partial \mathbf{B}_2}{\partial \mathbf{W}_1} \\
\frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{W}_2} &= \frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{L}_1} \frac{\partial \mathbf{L}_1}{\partial \mathbf{A}_1} \frac{\partial \mathbf{A}_1}{\partial \mathbf{B}_1} \frac{\partial \mathbf{B}_1}{\partial \mathbf{C}_1} \frac{\partial \mathbf{C}_1}{\partial \mathbf{W}_2} + \frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{A}_2} \frac{\partial \mathbf{A}_2}{\partial \mathbf{B}_2} \frac{\partial \mathbf{B}_2}{\partial \mathbf{W}_2} \\
\frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{W}_3} &= \frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{L}_1} \frac{\partial \mathbf{L}_1}{\partial \mathbf{A}_1} \frac{\partial \mathbf{A}_1}{\partial \mathbf{B}_1} \frac{\partial \mathbf{B}_1}{\partial \mathbf{C}_2} \frac{\partial \mathbf{C}_2}{\partial \mathbf{D}_2} \frac{\partial \mathbf{D}_2}{\partial \mathbf{E}_2} \frac{\partial \mathbf{E}_2}{\partial \mathbf{W}_3} + \frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{A}_2} \frac{\partial \mathbf{A}_2}{\partial \mathbf{B}_2} \frac{\partial \mathbf{B}_2}{\partial \mathbf{W}_3} \\
\frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{W}_4} &= \frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{L}_1} \frac{\partial \mathbf{L}_1}{\partial \mathbf{A}_1} \frac{\partial \mathbf{A}_1}{\partial \mathbf{B}_1} \frac{\partial \mathbf{B}_1}{\partial \mathbf{C}_2} \frac{\partial \mathbf{C}_2}{\partial \mathbf{W}_4} + \frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{A}_2} \frac{\partial \mathbf{A}_2}{\partial \mathbf{B}_2} \frac{\partial \mathbf{B}_2}{\partial \mathbf{W}_4} \\
\frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{b}_1} &= \frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{L}_1} \frac{\partial \mathbf{L}_1}{\partial \mathbf{A}_1} \frac{\partial \mathbf{A}_1}{\partial \mathbf{B}_1} \frac{\partial \mathbf{B}_1}{\partial \mathbf{C}_1} \frac{\partial \mathbf{C}_1}{\partial \mathbf{D}_1} \frac{\partial \mathbf{D}_1}{\partial \mathbf{b}_1} \\
\frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{b}_2} &= \frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{L}_1} \frac{\partial \mathbf{L}_1}{\partial \mathbf{A}_1} \frac{\partial \mathbf{A}_1}{\partial \mathbf{B}_1} \frac{\partial \mathbf{B}_1}{\partial \mathbf{b}_2} \\
\frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{W}_3} &= \frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{L}_1} \frac{\partial \mathbf{L}_1}{\partial \mathbf{A}_1} \frac{\partial \mathbf{A}_1}{\partial \mathbf{B}_1} \frac{\partial \mathbf{B}_1}{\partial \mathbf{C}_2} \frac{\partial \mathbf{C}_2}{\partial \mathbf{D}_2} \frac{\partial \mathbf{D}_2}{\partial \mathbf{b}_3} \\
\frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{W}_4} &= \frac{\partial \mathbf{L}(\boldsymbol{\Theta})}{\partial \mathbf{L}_1} \frac{\partial \mathbf{L}_1}{\partial \mathbf{A}_1} \frac{\partial \mathbf{A}_1}{\partial \mathbf{B}_1} \frac{\partial \mathbf{B}_1}{\partial \mathbf{b}_4}
\end{aligned}$$

2 Problem 2

For layer with input shape is $W_1 \times H_1 \times C$, suppose number of filters is K, the filter size is F, the stride is S, the zero padding is P. This will produce an output of $W_2 \times H_2 \times K$ where:

$$W_2 = (W_1 - F + 2P)/S + 1 \in R \Rightarrow W_1 - F + 2P(mod S) \equiv 0$$

$$H_2 = (H_1 - F + 2P)/S + 1 \in R \Rightarrow H_1 - F + 2P(mod S) \equiv 0$$

For max pooling, number of parameters

$$N = F^2CK + K.$$

For Convolution layer, number of parameters is 0.

For Conv 1, F = 5 , S = 3, K = 8, W1 = 100, H1 = 100, C = 3

$$\therefore 100 - 5 + 2P(mod 3) \equiv 0$$

$$\therefore P_{1min} = 2$$

$$\therefore W_2 = 34, H_2 = 34,$$

the shape of activation map is $34 \times 34 \times 8$; the number of parameters is 608.

For Maxpool 1, F = 2, S = 2, K = 8, W1 = 34, H1 = 34, C = 8,

$$\therefore 34 - 2 + 2P(mod 2) \equiv 0$$

$$\therefore P_{2min} = 0$$

$$\therefore W_2 = 17, H_2 = 17,$$

the shape of activation map is $17 \times 17 \times 8$; the number of parameters is 0.

For Conv 2, F = 3, S = 2, K = 16, W1 = 17, H1 = 17, C = 8

$$\therefore 17 - 3 + 2P(mod 2) \equiv 0$$

$$\therefore P_{3min} = 0$$

$$\therefore W_2 = 8, H_2 = 8,$$

The shape of activation map is $8 \times 8 \times 16$; the number of parameters is 1168.

For Maxpool 2, $F = 3$, $S = 3$, $K = 16$, $W_1 = 8$, $H_1 = 8$, $C = 16$

$$\because 8 - 3 + 2P(mod3) \equiv 0$$

$$\therefore P_{min} = 2$$

$$\therefore W_2 = 4, H_2 = 4,$$

the shape of activation map is $4 \times 4 \times 16$; the number of parameters is 0.

3 Problem 3

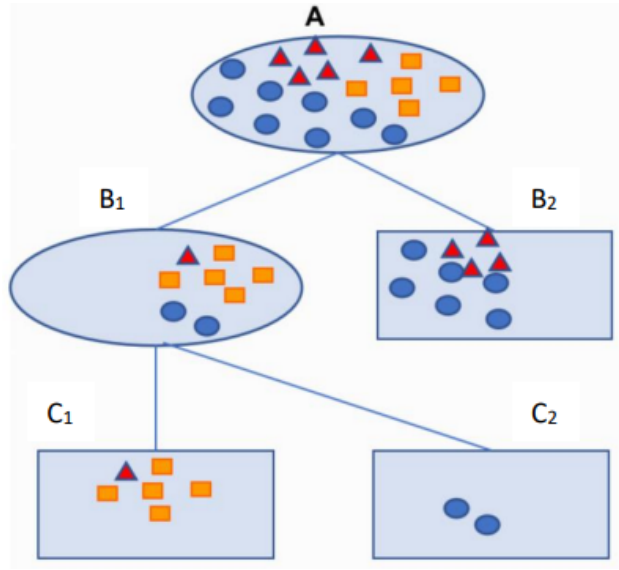


Figure 2: q3.

Node A:

$$entropy : -\frac{8}{18}\log_2\left(\frac{8}{18}\right) - \frac{5}{18}\log_2\left(\frac{5}{18}\right) - \frac{5}{18}\log_2\left(\frac{5}{18}\right) = 1.547$$

$$Gini\ index : \frac{8}{18}\left(1 - \frac{8}{18}\right) + \frac{5}{18}\left(1 - \frac{5}{18}\right) + \frac{5}{18}\left(1 - \frac{5}{18}\right) = 0.648$$

$$Misclassification\ Error : \frac{5}{9}$$

Node B1:

$$entropy : -\frac{1}{8}\log_2\left(\frac{1}{8}\right) - \frac{2}{8}\log_2\left(\frac{2}{8}\right) - \frac{5}{8}\log_2\left(\frac{5}{8}\right) = 1.299$$

$$Gini\ index : \frac{1}{8}\left(1 - \frac{1}{8}\right) + \frac{2}{8}\left(1 - \frac{2}{8}\right) + \frac{5}{8}\left(1 - \frac{5}{8}\right) = 0.531$$

$$\text{Misclassification Error} : \frac{3}{8}$$

Node B2:

$$\text{entropy} : -\frac{6}{10}\log_2\left(\frac{6}{10}\right) - \frac{4}{10}\log_2\left(\frac{4}{10}\right) = 0.971$$

$$\text{Gini index} : \frac{6}{10}\left(1 - \frac{6}{10}\right) + \frac{4}{10}\left(1 - \frac{4}{10}\right) = 0.48$$

$$\text{Misclassification Error} : \frac{2}{5}$$

Node C1:

$$\text{entropy} : -\frac{1}{6}\log_2\left(\frac{1}{6}\right) - \frac{5}{6}\log_2\left(\frac{5}{6}\right) = 0.65$$

$$\text{Gini index} : \frac{1}{6}\left(1 - \frac{1}{6}\right) + \frac{5}{6}\left(1 - \frac{5}{6}\right) = 0.278$$

$$\text{Misclassification Error} : \frac{1}{6}$$

Node C2:

$$\text{entropy} : 0$$

$$\text{Gini index} : 0$$

$$\text{Misclassification Error} : 0$$

4 Problem 4

4.1 Problem 4.a

Using the problem Hint, by calculation, you can easily get that:

empirical mean squared error (MSE) = 4.1, Bias² = 0.25, Variance = 4.01

$$\text{Average Prediction} = \frac{6 + 8 + 9 + 5 + 9 + 5 + 4 + 8 + 9 + 4}{10} = 6.7$$

$$\begin{aligned} \widehat{MSE} &= \frac{1}{10} [(6 - 7)^2 + (8 - 7)^2 + (9 - 7)^2 + (5 - 7)^2 + (9 - 7)^2 + (5 - 7)^2 \\ &\quad + (4 - 7)^2 + (8 - 7)^2 + (9 - 7)^2 + (4 - 7)^2] = 4.1 \end{aligned}$$

$$\begin{aligned} \text{Variance} &= \frac{1}{10} [(6 - 6.7)^2 + (8 - 6.7)^2 + (9 - 6.7)^2 + (5 - 6.7)^2 + (9 - 6.7)^2 \\ &\quad + (5 - 6.7)^2 + (4 - 6.7)^2 + (8 - 6.7)^2 + (9 - 6.7)^2 + (4 - 6.7)^2] = 4.01 \end{aligned}$$

$$\text{Bias}^2 = (7.2 - 6.7)^2 = 0.25$$

(1)

4.2 Problem 4.b

Explain why $\widehat{MSE}(x, y) \neq \text{Bias}^2 + \text{Variance} + \sigma^2$.

Assume the training dataset is D , and assume the learned hypothesis function (According to a certain algorithm $\mathcal{A}$) on D is h_D (You can regard the h_D as the function which accepts the input and outputs the corresponding prediction). From the Bias-Variance decomposition (You can refer to the lecture slides for detail):

By denoting the expected hypothesis function:

$$\bar{h} = E_{D \sim P^n} [h_D] = \int_D h_D p(D) dD \quad (2)$$

Please note that here a certain dataset D is the random variable to take the expectation. P^n is the joint distribution for n data samples: $(x, y) \sim P(\mathcal{X}, \mathcal{Y})$. By the decomposition calculation (Refer the lecture slides), finally, for expected test error, we can get that:

$$\begin{aligned} & E_{(\mathbf{x}, y), D} [(h_D(\mathbf{x}) - y)^2] \\ &= E_{(\mathbf{x}, y), D} [(h_D(\mathbf{x}) - \bar{h}(\mathbf{x}))^2] + E_{(\mathbf{x}, y)} [(\bar{h}(\mathbf{x}) - t(\mathbf{x}))^2] + E_{(\mathbf{x}, y)} [(t(\mathbf{x}) - y)^2] \end{aligned} \quad (3)$$

where the above three terms are variance, bias², noise, respectively. The key point is that, for the machine learning process, we only have and perform train/test on a certain dataset D . Hence we can not exactly get the P^n and then get the $\bar{h}$, hence with the assumption that our known data has the same distribution for the unknown data, we use the empirical mean squared error $\widehat{MSE}(x, y)$ to do the estimation. This is the reason why $\widehat{MSE}(x, y) \neq \text{Bias}^2 + \text{Variance} + \sigma^2$. In more detail, in this problem, we only sample 10 times and calculate the average:

$$\begin{aligned} \text{Variance} + \text{Bias}^2 + \sigma^2 &= 4.1 + 0.25 + 1 = 5.35 \\ \widehat{MSE}(x, y) &= 4.1 \\ \text{Variance} + \text{Bias}^2 + \sigma^2 &\neq \widehat{MSE} \end{aligned} \quad (4)$$

5 Problem 5

Method 1:

The relation between $\tanh(a)$ and $\sigma(a)$ is

$$\begin{aligned} \tanh(a) &= \frac{e^a - e^{-a}}{e^a + e^{-a}} \\ &= \frac{2e^a}{e^a + e^{-a}} - 1 \\ &= 2 \cdot \frac{1}{1 + e^{-2a}} - 1 \\ &= 2\sigma(2a) - 1 \end{aligned}$$

Replace $\tanh(a) = 2\sigma(2a) - 1$ into the two-layer network function with $\tanh(a)$ as hidden unit activation function, we have

$$\begin{aligned}
\hat{y}_k(\mathbf{x}, \hat{\mathbf{w}}) &= \sigma\left(\sum_{j=1}^M \hat{w}_{kj}^{(2)} \tanh\left(\sum_{i=1}^D \hat{w}_{ji}^{(1)} x_i + \hat{w}_{j0}^{(1)}\right) + \hat{w}_{k0}^{(2)}\right) \\
&= \sigma\left(\sum_{j=1}^M \hat{w}_{kj}^{(2)} \left[2\sigma\left(\sum_{i=1}^D 2\hat{w}_{ji}^{(1)} x_i + 2\hat{w}_{j0}^{(1)}\right) - 1\right] + \hat{w}_{k0}^{(2)}\right) \\
&= \sigma\left(\sum_{j=1}^M 2\hat{w}_{kj}^{(2)} \sigma\left(\sum_{i=1}^D 2\hat{w}_{ji}^{(1)} x_i + 2\hat{w}_{j0}^{(1)}\right) - \sum_{j=1}^M \hat{w}_{kj}^{(2)} + \hat{w}_{k0}^{(2)}\right)
\end{aligned}$$

Compared to

$$y_k(\mathbf{x}, \hat{\mathbf{w}}) = \sigma\left(\sum_{j=1}^M w_{kj}^{(2)} \sigma\left(\sum_{i=1}^D w_{ji}^{(1)} x_i + w_{j0}^{(1)}\right) + w_{k0}^{(2)}\right)$$

There exists linear transformation between w and $\hat{w}$, that enable $\forall \mathbf{x}, \mathbf{y}_k(\mathbf{x}, \hat{\mathbf{w}}) = \hat{\mathbf{y}}_k(\mathbf{x}, \hat{\mathbf{w}})$

$$\begin{aligned}
w_{ji}^{(1)} &= 2\hat{w}_{ji}^{(1)} & j = 1 \cdots M, i = 1 \cdots D \\
w_{j0}^{(1)} &= 2\hat{w}_{j0}^{(1)} & j = 1 \cdots M \\
w_{kj}^{(2)} &= 2\hat{w}_{kj}^{(2)} & j = 1 \cdots M \\
w_{k0}^{(2)} &= -\sum_{j=1}^M \hat{w}_{kj}^{(2)} + \hat{w}_{k0}^{(2)} & j = 1 \cdots M
\end{aligned}$$

Method 2:

The relation between $\tanh(a)$ and $\sigma(a)$ is

$$\begin{aligned}
\tanh(a) &= \frac{e^a - e^{-a}}{e^a + e^{-a}} \\
&= 1 - \frac{2e^{-a}}{e^a + e^{-a}} \\
&= 1 - 2 \cdot \frac{1}{e^{2a} + 1} \\
&= 1 - 2\sigma(-2a)
\end{aligned}$$

Replace $\tanh(a) = 1 - 2\sigma(-2a)$ into the two-layer network function with $\tanh(a)$ as

hidden unit activation function, we have

$$\begin{aligned}
\hat{y}_k(\mathbf{x}, \hat{\mathbf{w}}) &= \sigma\left(\sum_{j=1}^M \hat{w}_{kj}^{(2)} \tanh\left(\sum_{i=1}^D \hat{w}_{ji}^{(1)} x_i + \hat{w}_{j0}^{(1)}\right) + \hat{w}_{k0}^{(2)}\right) \\
&= \sigma\left(\sum_{j=1}^M \hat{w}_{kj}^{(2)} \left[1 - 2\sigma\left(\sum_{i=1}^D -2\hat{w}_{ji}^{(1)} x_i - 2\hat{w}_{j0}^{(1)}\right)\right] + \hat{w}_{k0}^{(2)}\right) \\
&= \sigma\left(\sum_{j=1}^M -2\hat{w}_{kj}^{(2)} \sigma\left(\sum_{i=1}^D -2\hat{w}_{ji}^{(1)} x_i - 2\hat{w}_{j0}^{(1)}\right) + \sum_{j=1}^M \hat{w}_{kj}^{(2)} + \hat{w}_{k0}^{(2)}\right)
\end{aligned}$$

Compared to

$$y_k(\mathbf{x}, \hat{\mathbf{w}}) = \sigma\left(\sum_{j=1}^M w_{kj}^{(2)} \sigma\left(\sum_{i=1}^D w_{ji}^{(1)} x_i + w_{j0}^{(1)}\right) + w_{k0}^{(2)}\right)$$

There exists linear transformation between w and $\hat{w}$, that enable $\forall \mathbf{x}, \mathbf{y}_k(\mathbf{x}, \hat{\mathbf{w}}) = \hat{\mathbf{y}}_k(\mathbf{x}, \hat{\mathbf{w}})$

$$\begin{aligned}
w_{ji}^{(1)} &= -2\hat{w}_{ji}^{(1)} & j = 1 \cdots M, i = 1 \cdots D \\
w_{j0}^{(1)} &= -2\hat{w}_{j0}^{(1)} & j = 1 \cdots M \\
w_{kj}^{(2)} &= -2\hat{w}_{kj}^{(2)} & j = 1 \cdots M \\
w_{k0}^{(2)} &= \sum_{j=1}^M \hat{w}_{kj}^{(2)} + \hat{w}_{k0}^{(2)} & j = 1 \cdots M
\end{aligned}$$